

Chiamanda Ononiwu

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EDUCATION

McNeese State University

Bachelor of Science in Electrical Engineering, minor in Computer Science; **GPA: 4.0**

Lake Charles, LA

Graduation date: May 2028

Awards: Dean's List 2024 – Present, McNeese Honors Scholar

Cornell University, Machine Learning Foundations e-Certificate, Break Through Tech

May 2025 – August 2025

TECHNICAL SKILLS

ML & AI: PyTorch, TensorFlow, Scikit-learn, CNNs, Vision Transformers, NLP, RAG pipelines, LLMs, Prompt Engineering

Hardware & EDA: KiCad (PCB schematic & layout), LTspice (AC/DC circuit simulation), Falstad, TinkerCAD, Fusion 360, ESP32, IoT sensors, Oscilloscope

Languages & Data: Python, C++, Pandas, NumPy, Scikit-learn, SQLite, Git

PROFESSIONAL EXPERIENCE

Ursa Space Systems, Break Through Tech AI fellow

August 2025 – December 2025

- Developed a **neural network classification pipeline** using PyTorch Vision Transformers on SAR imagery achieving **80% recall** on 1,400+ samples. Applied ML model development and debugging to ambiguous, high-variance data analogous to thermal and IR drop signal analysis.
- Engineered **EDA and feature extraction pipelines** (Python, Scikit-learn, SARPpy) that improved classification recall by **52.83%**, demonstrating iterative algorithm development and quantitative performance evaluation.
- Performed **comparative analysis across model architectures** to identify the most reliable model for real-world deployment; communicated findings through structured technical reporting.

CEAMLS 2025 Summer AI Research Institute, Undergraduate Research Project Manager

May 2025 – August 2025

- Architected a **full IoT data pipeline** integrating ESP32 sensor hardware with a Python ML backend for an AI-powered smart waste monitoring system; conducted literature reviews on IoT-ML systems to ground design decisions.
- Built **Random Forest predictive models** (Pandas, NumPy, Scikit-learn) projected to improve route efficiency by **40%**; **authored the methodology section** of a pending IEEE manuscript.

PROJECTS

Circuits Coursework (AC Circuit Analysis & Power Integrity)

Jan 2026 – May 2026

- Performed **AC power analysis** including real, reactive, and apparent power calculations across RLC networks, developing a foundational understanding of **power delivery and current distribution**.
- Simulated and analyzed **voltage drops, impedance mismatches, and frequency-domain behavior** using **LTspice**, gaining hands-on experience with the EDA simulation workflow Cadence tools extend to full-chip power integrity.

Digital Systems Design Lab

Aug 2025 – May 2026

- Designed **XOR/XNOR combinational circuits from NAND/NOR gates** using Boolean algebra and logic minimization, building deep familiarity with gate-level signal behavior and switching activity.
- Engineered **flip-flops from NAND gates** and constructed **synchronous counters**, analyzing **state transitions and timing behavior**, building foundational IR drop and dynamic power intuition.
- Optimized Boolean function realization by implementing decoder/multiplexer logic and Binary-to-Gray code conversion, resulting in a **systematic design-and-verify workflow** that reduced logic complexity.

StudyBot - AI RAG Study Assistant (CodePath) [GitHub](#)

April 2026

- Designed a **RAG-powered AI agent** using Gemini with engineered LLM guardrails to prevent hallucination; applied **prompt engineering** and pytest validation to ensure reliable, grounded outputs.

6 Key Macropad

December 2025 - Present

- Designed full PCB schematic and layout in KiCad** for a programmable input device, managing component placement, trace routing, and design rule checks; hence building the PCB/package design fundamentals Cadence's thermal and power analysis tools operate on.
- Programmed **embedded firmware on a microcontroller** and synchronized hardware with software for programmable keymaps, gaining end-to-end hardware–software integration experience.

LEADERSHIP EXPERIENCE

McNeese NSBE, TORCH Chair

April 2025 – May 2026

- Organized a community **STEM academic tech bowl serving 50+ attendees**, collaborating with Louisiana Region V STEM center and managing on-site logistics.
- Built a centralized database of **15+ STEM outreach partners** across Calcasieu Parish, creating new partnerships that enabled members to deliver STEM lessons to K–12 students.
- Organized trips to different high schools to talk about engineering and college life, leading to some kids deciding to pursue engineering.